

generalized nondeterministic finit automaton (gnfa)

a gnfa (N) of a dfa (M) is a special type where $L(N) = L(M)$, where N has exactly one accept state, and all transitions of N are regular expressions.

example:

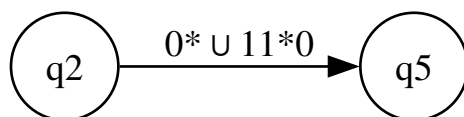


Figure 1: graphs/06_gnfa.gv

at q_2 , if you encounter a string in the language $0^* \cup 11^*0$, move to state q_5 .

$$0^* \cup 11^*0 = \{\epsilon, 0, 00, \dots\} \cup \{10, 110, 1110, \dots\}$$

converting dfa to gnfa

to convert a dfa to a gnfa, we need to:

1. start with a dfa M
2. add a new start state s with an ϵ transition to the original start state of M
3. add a new accept state a
 - from every accept state of M , add an ϵ arrow to the new accept state
 - change all original accept states of M to non-accept states
4. turn transition labels to regular expressions
5. add necessary transition arrows
 - change multiple arrows or multiple labels to a single arrow while label is the union of the previous labels, like this:

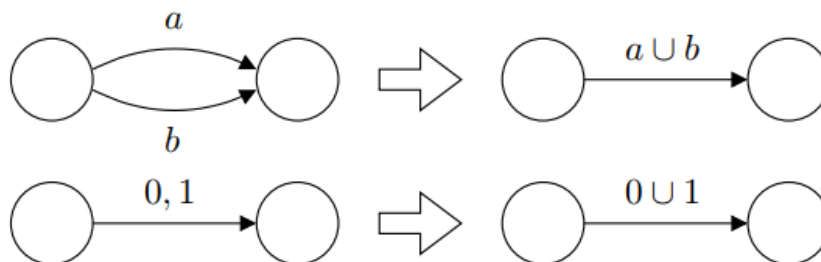


Figure 2: dfa to gnfa arrows